Question:

There exist real numbers x and y, both greater than 1, such that $\log_x\left(y^x\right)=\log_y\left(x^{4y}\right)=10$. Find xy.

Reasoning + Answer:

<think> To solve for \(x\) and \(y\) such that \(\log_x\left(y^x\right)=\log_y\left(x^{4y}\right)=10\), we start by using the properties of logarithms. The first equation can be rewritten as:

\[
\log_x\left(y^x\right) = x \log_x y = 10
\]

Similarly, the second equation can be rewritten as:

\[
\log_y\left(x^{4y}\right) = 4y \log_y x = 10
\]

From these, we have:

\[
\log_x y = \frac{10}{x} \quad \text{and} \quad \log_y x = \frac{10}{4y} = \frac{5}{2y}
\]

Using the change of base formula for logarithms, \(\log_x y = \frac{1}{\log_y x}\), we substitute \(\log_x y\) from the first equation into the second equation:

\[
\frac{10}{x} = \frac{2y}{5}
\]

This simplifies to:

\[
50 = 2xy \implies xy = 25
\]

Therefore, the value of \(xy\) is:

\[
\boxed{25}
\]

<answer>25</answer>